INCH-POUND

MS75085E
W/AMENDMENT 1
7 October 2022
SUPERSEDING
MS75085E
31 August 2007

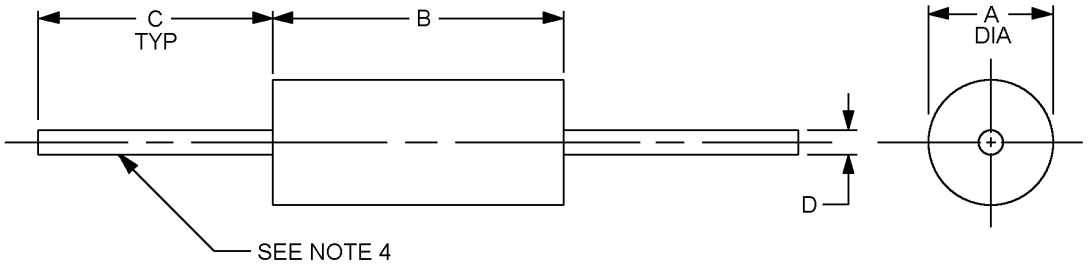
MILITARY SPECIFICATION SHEET

# COILS, RADIO FREQUENCY, MOLDED, FIXED, (FERRITE CORE), TYPES LT10K078 TO LT10K096, INCLUSIVE

Inactive for new design, after 4 September 1985.
For new design, use MIL-PRF-39010/10.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-15305.




| LTR | Minimum | Maximum |
|---|---|---|
| A | .085 (2.16) | .105 (2.67) |
| B | .240 (6.10) | .260 (6.60) |
| C | 1.250 (31.75) | 1.625 (41.28) |
| D | .0185 (0.470) | .0215 (0.546) |

NOTES:
1. Dimensions are in inches with metric equivalents (mm) in parentheses.
2. Metric equivalents are given for general information only.
3. These coils are intended to be supported by their leads.
4. Solderable/weldable lead wire, number 24 AWG.

FIGURE 1. Dimensions and configuration.


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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Weight: 0.0106 ounce, maximum.

Operating temperature range: -55° to +105°C.

Ambient temperature: + 90°C maximum.

Temperature rise: 15°C.

Terminal pull: 5 pounds, minimum.

Altitude: 70,000 feet.

Shock, specified pulse: MIL-STD-202-213, test condition I, is applicable.

Dielectric withstanding voltage:

At sea level: MIL-STD-202-301, test voltage 1,000 V rms for a minimum of 60 seconds.

At reduced barometric pressure: MIL-STD-202-105, test condition C, test voltage 200 V rms for a minimum of 60 seconds.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Q values: See table I.

Self-resonant frequency (SRF): See table I.

DC resistance (DCR): See table I.

Part or Identifying Number (PIN): MS75085 - (dash number from table I).



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TABLE 1. Electrical characteristics (initial).

| Dash Number MS75085 | Type Designation | Inductance (μH) ±10% | Q Min. | Test Frequency (L and Q) (MHz) | SRF Minimum (MHz) | DC resistance (25°C) max (Ohms) | Rated DC current, (mA) |
|---|---|---|---|---|---|---|---|
| -1 | LT10K078 | 33 | 45 | 2.5 | 24 | 3.4 | 130 |
| -2 | LT10K079 | 39 | 45 | 2.5 | 22 | 3.6 | 125 |
| -3 | LT10K080 | 47 | 45 | 2.5 | 20 | 4.5 | 110 |
| -4 | LT10K081 | 56 | 45 | 2.5 | 18 | 5.7 | 100 |
| -5 | LT10K082 | 68 | 50 | 2.5 | 15 | 6.7 | 92 |
| -6 | LT10K083 | 82 | 50 | 2.5 | 14 | 7.3 | 88 |
| -7 | LT10K084 | 100 | 50 | 2.5 | 13 | 8 | 84 |
| -8 | LT10K085 | 120 | 30 | .79 | 12 | 13 | 66 |
| -9 | LT10K086 | 150 | 30 | .79 | 11 | 15 | 61 |
| -10 | LT10K087 | 180 | 30 | .79 | 10 | 17 | 57 |
| -11 | LT10K088 | 220 | 30 | .79 | 9 | 21 | 52 |
| -12 | LT10K089 | 270 | 30 | .79 | 8 | 25 | 47 |
| -13 | LT10K090 | 330 | 30 | .79 | 7 | 28 | 45 |
| -14 | LT10K091 | 390 | 30 | .79 | 6.5 | 35 | 40 |
| -15 | LT10K092 | 470 | 30 | .79 | 6 | 42 | 36 |
| -16 | LT10K093 | 560 | 30 | .79 | 5 | 46 | 35 |
| -17 | LT10K094 | 680 | 30 | .79 | 4 | 60 | 30 |
| -18 | LT10K095 | 820 | 30 | .79 | 3.8 | 65 | 29 |
| -19 | LT10K096 | 1000 | 30 | .79 | 3.4 | 72 | 28 |

TABLE II. Electrical characteristics (final). 1/

| Inspection group | Allowable variation from Initial measurement | Allowable variation from Initial measurement | Allowable percent from specified minimum value in electrical characteristics (initial) table | Allowable percent from specified minimum value in electrical characteristics (initial) table |
|---|---|---|---|---|
| | Inductance (percent) | DC resistance | Self-resonant frequency | Q |
| Qualification inspection | | | | |
| Group II | ±2 | --- | --- | -10 |
| Group III | ±10 | ±(5% +.001 ohm) | 15 | -20 |
| Group IV | ±5 | ±(2% +.001 ohm) | -10 | -15 |
| Conformance inspection group C | | | | |
| Subgroup I | ±2 | --- | --- | -10 |
| Subgroup II | ±5 | ±(2% +.001 ohm) | -10 | -15 |
| Subgroup III | ±10 | ±(5% +.001 ohm) | -15 | -20 |

1/ Test fixture allowance of +.01μH shall be added to all change in inductance limits ±( _ percent +.01 μH).



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Application notes:

1. After the overload test is performed, a period of 24 hours shall elapse prior to taking electrical characteristics (final) measurements.
2. The polarization voltage during moisture the moisture resistance tests is applied with the positive lead connected to the coil terminals tied together, and the negative lead connected to the metal strap.
3. DC resistance shall be the last measurement taken in the electrical characteristics test sequence.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-15305, this document references the following:

MIL-STD-202-105 MIL-STD-202-213 MIL-STD-202-301

Custodians:
Army – CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA – CC

Project 5950-2022-065

Review activities:
Army – AR, CR4, MI
Navy – AS, CG, MC, OS, SH
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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